Google Sheets Cheat Sheet

Key Terms & Keywords

Common Spreadsheet Concepts

- **Spreadsheet:** A grid of rows and columns used for data storage and analysis.
- **Cell:** The intersection of a row and a column (e.g., A1, B5).
- **Formula:** A user-defined calculation starting with an equals sign (e.g., =A1+B1).
- **Function:** Pre-built calculations for performing specific tasks (e.g., SUM, AVERAGE).
- **Range:** A selection of multiple cells (e.g., A1:A10).
- **Sheet Tab:** The tabs at the bottom of Google Sheets that separate different worksheets.
- **Filter:** A tool to view specific rows that meet certain conditions.
- **Pivot Table:** A feature that summarizes large sets of data.

Commonly Used Formulas and Functions

1. Basic Formulas

- **Addition:** `=A1+B1`
- **Subtraction:** `=A1-B1`
- **Multiplication:** `=A1*B1`
- **Division:** `=A1/B1`

2. Basic Functions

- **SUM:** Adds a range of numbers.
  =SUM(A1:A10)
- **AVERAGE:** Calculates the average (mean) of a range.
  =AVERAGE(A1:A10)
- **COUNT:** Counts the number of numeric entries in a range.
  =COUNT(A1:A10)
- **COUNTA:** Counts all non-empty cells.
  =COUNTA(A1:A10)

3. Data Manipulation Functions

- **IF:** Returns a value based on a logical test.
  =IF(A1>10, "Yes", "No")
- **VLOOKUP:** Searches for a value in the first column of a range and returns a value in the same row from another column.
  =VLOOKUP(1001, A2:D10, 3, FALSE)
- **HLOOKUP:** Similar to VLOOKUP but searches for values in the top row.
  =HLOOKUP("Product", A1:D5, 2, FALSE)
- **INDEX:** Returns the value of a cell at the intersection of a specified row and column.
  =INDEX(A1:C10, 2, 3)
- **MATCH:** Returns the relative position of a value in a range.
  =MATCH(50, A1:A100, 0)
- **CONCAT:** Combines text from multiple cells.
  =CONCAT(A1, B1)
- **TEXTJOIN:** Joins text from multiple cells with a specified delimiter.
  =TEXTJOIN(", ", TRUE, A1:A5)

4. Data Analysis Functions

- **LEN:** Returns the length of a text string.
  =LEN(A1)
- **TRIM:** Removes extra spaces from text.
  =TRIM(A1)
- **SORT:** Sorts a range based on specified columns.
  =SORT(A1:B10, 1, TRUE)
- **FILTER:** Returns only the rows that meet specific criteria.
  =FILTER(A1:A10, B1:B10>50)
- **UNIQUE:** Returns unique values from a range.
  =UNIQUE(A1:A10)

5. Date and Time Functions

- **TODAY:** Returns the current date.
  =TODAY()
- **NOW:** Returns the current date and time.
  =NOW()
- **DATEDIF:** Calculates the difference between two dates.
  =DATEDIF(A1, B1, "D")
    - "D" for days, "M" for months, "Y" for years.

6. Logical Functions

- **IF:** Returns a value based on a logical test.
    - **Formula:** `=IF(condition, value_if_true, value_if_false)`
    - **Example:** `=IF(A1>100, "High", "Low")`

- **IFS:** Evaluates multiple conditions and returns the value for the first true condition.
  - **Formula:** `=IFS(condition1, value1, condition2, value2, ...)`
  - **Example:** `=IFS(A1>90, "A", A1>80, "B", A1>70, "C")`
- **SWITCH:** Returns a value based on matching cases.
  - **Formula:** `=SWITCH(expression, case1, value1, case2, value2, ..., default)`
  - **Example:** `=SWITCH(A1, 1, "One", 2, "Two", 3, "Three", "Other")`

7. Financial Functions

- **NPV (Net Present Value):** Calculates the net present value of an investment.
  - **Formula:** `=NPV(discount_rate, cash_flows)`
  - **Example:** `=NPV(0.05, B1:B5)`
- **IRR (Internal Rate of Return):** Calculates the internal rate of return for a series of cash flows.
  - **Formula:** `=IRR(cash_flows)`
  - **Example:** `=IRR(B1:B5)`

8. Array Formulas

- **ARRAYFORMULA:** Applies a formula to an entire range.
  - **Formula:** `=ARRAYFORMULA(array_formula)`
  - **Example:** `=ARRAYFORMULA(A1:A10 * B1:B10)`
- **FILTER:** Returns rows that meet specific criteria.
  - **Formula:** `=FILTER(range, condition1, [condition2, ...])`
  - **Example:** `=FILTER(A1:A10, B1:B10>50)`

9. Data Analysis & Visualization Functions

- **Pivot Tables:** Summarize large datasets.
  - Navigate to **Data > Pivot Table** and customize rows, columns, and values.
- **QUERY Function:** Perform SQL-like queries on data.
  - **Formula:** `=QUERY(data, query, [headers])`
  - **Example:** `=QUERY(A1:D100, "SELECT A, SUM(B) WHERE C > 100 GROUP BY A ORDER BY SUM(B) DESC")`
- **Charts:** Visualize data with various chart types (bar, line, pie, etc.).
  - Insert via **Insert > Chart** and customize ranges.

10. Data Validation

- **Drop-Down Lists:** Create structured input.
  - Navigate to **Data > Data Validation > List of Items**.

- **Custom Validation Rules:** Set rules like numeric limits or date ranges.
    - Example Formula for Validation: `=A1>0`
- **Conditional Data Validation:** Use formulas to create dynamic rules.
    - Example Formula: `=AND(A1>0, A1<100)`

Quick Keyboard Shortcuts

- **Select Entire Row:** `Shift + Space`
- **Select Entire Column:** `Ctrl + Space`
- **Insert Row:** `Ctrl + Shift + =`
- **Delete Row:** `Ctrl + -`
- **Open Format Menu:** `Ctrl + 1`
- **Undo:** `Ctrl + Z`
- **Redo:** `Ctrl + Y`
- **Find:** `Ctrl + F`
- **Create New Sheet:** `Shift + F11`

Tips for Maximising Efficiency

- Use **Conditional Formatting** to highlight important data based on conditions.
- Leverage **Data Validation** to create dropdown menus and prevent invalid entries.
- Utilise **Pivot Tables** to summarise and analyse large datasets quickly.
- Automate repetitive tasks with **Macros** and explore **Add-ons** for additional functionality.
